

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Pharm. Examination

University Theory Examination November-December 2015

PH219 Pharmaceutical Engineering-III

Date: 03/12/2015 , Thursday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Which of following is not a mechanism of size reduction
[A] Elutriation
[B] Cutting
[C] Impact
[D] Compression
 2. Elutriation depends upon the movement of a fluid _____ the direction of flow
[A] in the same
[B] perpendicular
[C] against
[D] all
 3. Size reduction is used to improve the
[A] Physical stability
[B] Dissolution ratre
[C] Rate of absorption
[D] All of above
 4. The number mentioned for the sieve is the number of meshes (Holes) in a length of
[A] 26.4 mm
[B] 24.4 mm
[C] 25.4 mm
[D] 23.4 mm
 5. Which of the following is not a size separation method
[A] Elutriation
[B] Sieving
[C] Ball Mill
[D] Cyclone Separator
 6. Mechanism involved in mixing of solid is
[A] Convection
[B] Diffusion
[C] Shear
[D] All

7. Crystal is a
 - [A] highly organized matter
 - [B] nonliving matter
 - [C] Both A and B
 - [D] None
8. Substance exist in more than one crystalline form is called as
 - [A] polymorphs
 - [B] Mumorphs
 - [C] Paramorphs
 - [D] Decamorphs
9. Swenson Walker Crystallizer works on a principle of
 - [A] evaporation
 - [B] salting out
 - [C] cooling
 - [D] semi evap
10. Mixing is a process to
 - [A] get homogeneous mixture
 - [B] promote physical change
 - [C] promote chemical reaction
 - [D] All
11. Working principle of ball mill is
 - [A] Impact
 - [B] Attrition
 - [C] Both A and B
 - [D] None
12. Range of particle size separated by cyclone separator is
 - [A] 2-50 μm
 - [B] 1-100 μm
 - [C] 50-100 μm
 - [D] 100-200 μm
13. Which of the following method is suitable for grinding of material having water content more than 50%
 - [A] Wet grinding
 - [B] Dry grinding
 - [C] Cutting
 - [D] Pressure Grinding
14. Most commonly used metal for making the sieve
 - [A] Zinc
 - [B] Aluminium
 - [C] Stainless Steel
 - [D] Tin
15. From the following which is not a shape of tumbler mixer?
 - [A] Double Cone
 - [B] Cubicle
 - [C] Y Shaped
 - [D] None
16. Turbine mixer is a example of
 - [A] solid mixer
 - [B] gas mixer
 - [C] semisolid mixer
 - [D] liquid mixer

17. From the following which is the type of fire detector?
 [A] UV/IR
 [B] PDA
 [C] Electric
 [D] Water
18. which of the following is not a type of fire extinguisher
 [A] CO₂
 [B] Dry Chemical
 [C] Halon
 [D] O₂
19. Hazards are potentially
 [A] errors
 [B] accidents
 [C] risks
 [D] traumas
20. which of the following are not a type of mechanical injury
 [A] crushing
 [B] cutting
 [C] burning
 [D] tearing

SECTION – II

- Q 2** Attempt any **FOUR** of the following
- | | | |
|----------|--|-----------|
| A | Describe Mier's supersaturation theory. What are its limitations? | 05 |
| B | Enumerate the continuous process crystallizers. Explain in detail swenson-walker crystallizer. | 05 |
| C | Write a note on Preventive measures for Mechanical Hazards. | 05 |
| D | Enumerate the methods of size separation and explain cyclone separator. | 05 |
| E | Write note on standards for sieve. | 05 |
| F | Write note on applications of size separation in pharmacy. | 05 |

SECTION – III

- Q 3** Attempt any **FOUR** of the following
- | | | |
|----------|--|-----------|
| A | Define the size reduction. Explain the mechanism of size reduction in detail. | 05 |
| B | Briefly explain the key factors for the selection of size reduction equipments. | 05 |
| C | Write a note on - Ball mill - construction and working. | 05 |
| D | Give application of size reduction in pharmaceutical engineering. Enlist the newer mills used in industries. | 05 |
| E | Write a note on hammer mill with possible variants. | 05 |
| F | What are the elements of automatic process control systems? Explain briefly. | 05 |
- Q 4** Attempt any **FOUR** of the following
- | | | |
|----------|--|-----------|
| A | Give the difference between the propeller mixer and turbine mixer. | 05 |
| B | Give the classification of mixer. | 05 |
| C | Explain any four factors that influencing the mixing. | 05 |
| D | Write a note on mixer used in semisolids. | 05 |
| E | Explain silversion mixer in detail. | 05 |
| F | Explain any one method to control humidity in pharmaceuticals. | 05 |